



smelt-ml: A Pure-Rust Machine Learning Framework Combining Spatial Modeling, Conformal Prediction, and Cache-Optimized Gradient Boosting

Francisco Parra

Universidad de Santiago de Chile

Abstract

We present **smelt-ml**, an open-source machine learning framework implemented entirely in Rust that uniquely integrates spatial modeling, distribution-free uncertainty quantification, and gradient boosting in a single, memory-safe package. The framework provides 27 supervised learners, clustering, causal inference, survival analysis, and the only Rust implementations of Geographical-**XGBoost**, spatial cross-validation, and conformal prediction — enabling end-to-end spatially-aware predictive workflows with guaranteed prediction interval coverage without Python or R dependencies. The gradient boosting implementations employ cache-optimized column-major histogram accumulation with **u8** packing. On single-threaded benchmarks (n from 500 to 100,000, 10 runs, mean $\pm$ std), **smelt-ml** is 1.4–8.3 $\times$ faster than official C++ libraries at research-typical scales ($n \leq 10,000$) and consumes an order of magnitude less memory, while C++ is faster at $n \geq 50,000$. The composable, trait-based architecture is inspired by **mlr3** (Lang, Binder, Richter, Schratz, Pfisterer, Coors, Au, Casalicchio, Kotthoff, and Bischl 2019) and **scikit-learn** (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot, and Duchesnay 2011), with zero **unsafe** blocks across 13,000+ lines of code.

Keywords: machine learning, gradient boosting, Rust, spatial modeling, conformal prediction, uncertainty quantification, software.

1. Introduction

Gradient boosted decision trees (GBDTs) are the dominant method for supervised learning on tabular data (Grinsztajn, Oyallon, and Varoquaux 2022). The three most widely used implementations — **XGBoost** (Chen and Guestrin 2016), **LightGBM** (Ke, Meng, Finley, Wang,

Chen, Ma, Ye, and Liu 2017), and **CatBoost** (Prokhorenkova, Gusev, Vorobev, Dorogush, and Gulin 2018) — are implemented in C++ and have been optimized over many years. These libraries serve as the de facto baselines in the majority of applied machine learning studies.

Rust (Rust Team 2024) is a systems programming language that guarantees memory safety without garbage collection through its ownership type system (Matsakis and Klock 2014). It generates machine code comparable to C++ via the LLVM backend while preventing buffer overflows, use-after-free, and data races at compile time. Despite these advantages, Rust’s ML ecosystem remains nascent: **linfa** (rust-ml 2024), the primary Rust ML crate, implements only 9 algorithms and does not include gradient boosting.

Beyond the language gap, a practical gap exists in the spatial ML workflow: practitioners who need spatially-aware prediction with uncertainty quantification must currently combine multiple Python or R packages with incompatible APIs (e.g., **scikit-learn** for modeling, **geoxgboost** for spatial regression, **MAPIE** for conformal intervals, and custom spatial resampling code). No existing framework in any language integrates all of these in a single composable pipeline.

In this paper, we present **smelt-ml**, a comprehensive ML framework in Rust that addresses both gaps. Our main contributions are:

1. An **integrated spatial ML pipeline** that uniquely combines Geographical-**XGBoost** (Grekousis 2025), spatial cross-validation, and conformal prediction (Romano, Patterson, and Candès 2019) in a single composable framework — enabling spatially-aware modeling with distribution-free uncertainty quantification without external dependencies.
2. **Cache-optimized gradient boosting** using column-major bin storage with u8 packing, achieving performance competitive with official C++ libraries at research-typical scales ($n \leq 10,000$) and an order-of-magnitude reduction in memory consumption.
3. A **comprehensive ML framework** with 27 supervised learners, clustering, survival analysis, Causal Forest (Wager and Athey 2018), Oblique Forest (Tomita, Browne, Shen, Chung, Patsolic, Falk, Priebe, Yim, Burns, Maggioni, and Vogelstein 2020), and streaming Hoeffding Trees — the most complete ML framework in Rust.
4. **Validation** showing accuracy comparable to **scikit-learn** across standard benchmarks, with zero **unsafe** blocks across the entire codebase.

2. Software architecture

2.1. Design principles

smelt-ml follows a trait-based architecture inspired by **mlr3**:

```
Data (CSV) -> Task -> Pipeline(Transformers -> Learner)
               -> Prediction -> Measure
```

The core abstractions are defined as Rust traits:

- **Learner**: Algorithm that trains on a **Task** and produces a **TrainedModel**. Requires **Send** + **Sync** for thread safety.
- **TrainedModel**: Fitted model that predicts on new data.
- **Measure**: Evaluation metric (Accuracy, F1, RMSE, R^2 , AUC-ROC, etc.).
- **Transformer**: Preprocessing step composable via **Pipeline**.
- **Resample**: Data splitting strategy.

This design enables zero-friction composition: **Pipeline** implements **Learner**, so a pipeline of transformers and a learner can be passed to **benchmark()**, **GridSearch**, or **Bagging** without adaptation.

2.2. Quick start

A minimal classification example:

```
use smelt_ml::prelude::*;
use ndarray::array;

let features = array![[0.0, 0.0], [1.0, 1.0],
                      [0.0, 1.0], [1.0, 0.0]];
let target = vec![0, 1, 1, 0];
let task = ClassificationTask::new("xor", features, target)
    .unwrap();

let mut xgb = XGBoost::new()
    .with_n_estimators(50)
    .with_max_depth(3);
let model = xgb.train_classif(&task).unwrap();
let pred = model.predict(task.features()).unwrap();
```

Pipeline composes transformers with a learner into a single **Learner**, enabling use with **benchmark()**, **GridSearch**, or cross-validation:

```
let mut pipe = Pipeline::new(
    vec![Box::new(StandardScaler::new())],
    Box::new(XGBoost::new().with_n_estimators(100)),
);
let model = pipe.train_classif(&task).unwrap();
```

2.3. Implemented algorithms

Table 1 summarizes the 27 supervised learners. Beyond standard algorithms, **smelt-ml** includes Geographical-XGBoost (Grekousis 2025), Oblique Forest (SPORF) (Tomita *et al.*

Category	Algorithm	Classif.	Regress.
Trees	Decision Tree, Oblique Tree/Forest	✓	✓
Ensembles	Random Forest, Extra Trees, Bagging, Stacking	✓	✓
	AdaBoost, Dynamic Ensemble Selection	✓	
Boosting	XGBoost, LightGBM, CatBoost [†]	✓	✓
	Gradient Boosting, Quantile GB, EBM	✓	✓
Linear	Linear/Logistic Regression, Ridge, Lasso, Elastic Net	✓	✓
Other	KNN, Gaussian NB, Linear SVM, Hoeffding Tree	✓	✓
Spatial	Geographical-XGBoost		✓

Table 1: Supervised learners in **smelt-ml**. [†]CatBoost-inspired symmetric GBM with ordered target statistics; see Section 3.4 for scope.

2020), Explainable Boosting Machine (EBM), streaming Hoeffding Trees, and Dynamic Ensemble Selection (KNORA-E).

Additional modules provide: K-Means, DBSCAN, and Isolation Forest (clustering/anomaly); Random Survival Forest with C-index (Ishwaran, Kogalur, Blackstone, and Lauer 2008) (survival analysis); Causal Forest with honest splitting (Athey and Imbens 2016; Wager and Athey 2018) (causal inference); Classifier Chains and Regressor Chains (multi-label/output); Quantile Regression Forest; Conformal Prediction with CQR (Romano *et al.* 2019); and SHAP-based feature importance.

2.4. Comparison with existing frameworks

Table 2 compares **smelt-ml** with **linfa** 0.7 (rust-ml 2024) (the primary Rust ML crate), **scikit-learn** (Pedregosa *et al.* 2011), and **mlr3** (Lang *et al.* 2019).

3. Cache-optimized gradient boosting

3.1. The histogram bottleneck

Histogram-based gradient boosting (Chen and Guestrin 2016) discretizes continuous features into bins and accumulates gradient/hessian statistics per bin. The key computational kernel is:

$$\text{hist}[b] += g_i, \quad b = \text{bin_index}[i][\text{feature}] \quad (1)$$

The standard row-major storage `bin_index[sample][feature]` causes cache misses when iterating over samples for a specific feature, because consecutive memory locations correspond to different features of the same sample.

3.2. Column-major storage with u8 packing

Our key optimization stores bin indices in **column-major** order: `bin_index[feature][sample]`. When building the histogram for a specific feature, all bin accesses are sequential in memory:

Category	smelt-ml	linfa	sklearn	mlr3
Gradient Boosting	✓ ^a	—	✓ ^b	✓ ^b
Random Forest / Extra Trees	✓	—	✓	✓
Oblique Forest (SPORF)	✓	—	—	—
Decision Tree	✓	✓	✓	✓
Linear / Logistic / Ridge	✓	✓	✓	✓
KNN / Naive Bayes	✓	✓	✓	✓
SVM (kernel)	Linear	✓	✓	✓
Neural Networks	—	—	✓	✓ ^c
K-Means / DBSCAN	✓	✓	✓	✓
PCA	✓	✓	✓	✓
Causal Forest	✓	—	—	— ^d
Survival Forest	✓	—	—	✓ ^c
Conformal Prediction	✓	—	—	—
Geographical-XGBoost	✓	—	—	—
Spatial cross-validation	✓	—	—	✓ ^c
SMOTE / ADASYN	✓	—	✓ ^e	✓ ^c
Bayesian tuning	✓	—	✓ ^f	✓
Language	Rust	Rust	Python	R
Supervised learners	27	9	40+	30+ ^c
unsafe blocks	0	—	N/A	N/A

Table 2: Feature comparison across ML frameworks. ^aFrom-scratch implementations. ^bVia external C++ libraries (XGBoost, LightGBM). ^cVia extension packages. ^dAvailable in R via **grf**. ^eVia **imbalanced-learn**. ^fVia **scikit-optimize** or **Optuna**.

```
// Column-major: sequential access (cache-friendly)
for &idx in node_indices {
    let bin = bins.get_bin(feature, idx);
    bin_g[bin] += gradients[idx];
    bin_h[bin] += hessians[idx];
}
```

Combined with **u8 packing** (254 real bins + 1 NaN sentinel fit in 1 byte vs. 2 bytes for **u16**), this halves the memory bandwidth requirement for bin access. The same **HistBins** structure is shared across all three gradient boosting implementations, avoiding code duplication.

3.3. Additional optimizations

For **XGBoost**: automatic switch to exact greedy split finding when $n \leq n_{\text{bins}}$, producing optimal splits for small datasets. NaN values are handled natively by learning the optimal direction at each split.

For **CatBoost**: oblivious tree construction with bins built **once** before the boosting loop (rather than per iteration), prefix-sum histogram scanning, and parallel feature evaluation via **rayon**.

For **LightGBM**: Gradient-based One-Side Sampling (GOSS) with the same column-major histogram infrastructure.

3.4. Scope of the CatBoost implementation

Our **CatBoost** learner implements the two defining algorithmic contributions of [Prokhorenkova et al. \(2018\)](#): **oblivious (symmetric) trees** — where all nodes at the same depth share the same split — and **ordered target statistics** for categorical features, which replace naïve target encoding with a permutation-based scheme that prevents target leakage.

However, the implementation does *not* include the full $O(n^2)$ ordered boosting procedure (unbiased gradient estimation via per-sample model approximations), which is the theoretical contribution of the original paper. Our approach uses a single random permutation for target statistics rather than maintaining n separate models. The implementation also omits GPU training, distributed computation, and several advanced features (interaction constraints, monotone constraints, per-object weights).

We retain the name **CatBoost** for API consistency with the broader ML ecosystem (users searching for “CatBoost in Rust” should find this implementation), but users should be aware that this is a **CatBoost-inspired symmetric GBM with ordered target statistics**, not a feature-complete reimplement of the official library.

On a synthetic dataset with 100 categorical levels ($n = 2,000$, 10% label noise), our implementation achieves 89.7% accuracy vs. 89.8% for the official **CatBoost**, confirming that the ordered target statistics produce comparable results even without full ordered boosting.

3.5. Scope of the LightGBM implementation

Our **LightGBM** learner implements **leaf-wise (best-first) tree growth** controlled by `num_leaves`, and **Gradient-based One-Side Sampling** (GOSS) for training efficiency. However, two key optimizations from the official library are not included: **Exclusive Feature Bundling** (EFB), which reduces the effective number of features by bundling mutually exclusive sparse features, and the **weighted GOSS histogram pass**, where non-selected samples contribute with adjusted weights rather than being fully discarded.

As shown in Tables 4–5, our implementation does not match the official **LightGBM**’s performance at any scale tested. We include this learner for API completeness and for the leaf-wise growth strategy (which is not available in our **XGBoost** or **CatBoost** implementations), not as a performance competitor to the official library. Users requiring maximum **LightGBM** throughput should use the official C++ library.

3.6. Feature comparison with official libraries

Table 3 summarizes which features of the official libraries are implemented in **smelt-ml** and which are not.

4. Performance evaluation

4.1. Experimental setup

Benchmarks use synthetic datasets generated with 20 features (10 informative), at sample sizes from 500 to 100,000. Configuration: 100 trees, max depth 6, learning rate 0.3 (**XGBoost/CatBoost**) or 0.1 (**LightGBM**). All benchmarks use **single-threaded execution** to

Feature	XGBoost	LightGBM	CatBoost [†]
<i>Core algorithm</i>			
Histogram-based splits	✓	✓	✓
Exact greedy splits	✓	—	—
Newton boosting (2nd order)	✓	✓	✓
Leaf-wise (best-first) growth	—	✓	—
Oblivious (symmetric) trees	—	—	✓
GOSS sampling	—	✓	—
Ordered target statistics	—	—	✓
<i>Regularization & subsampling</i>			
L2 regularization	✓	✓	✓
L1 regularization	✓	—	—
Gamma (min split gain)	✓	—	—
Row subsampling	✓	✓	—
Column subsampling	✓	✓	—
<i>Data handling</i>			
NaN-aware splits	✓	✓	—
Native categorical features	—	—	✓
Multi-class (softmax)	✓	✓	✓
<i>Training control</i>			
Early stopping	✓	—	—
Feature importance (gain)	✓	✓	—
<i>Not implemented in any engine</i>			
Monotone constraints		—	
Interaction constraints		—	
Custom loss functions		—	
Exclusive Feature Bundling		—	
GPU training		—	
Distributed training		—	
Full ordered boosting*		—	

Table 3: Implemented features in **smelt-ml**’s gradient boosting engines. ✓ = implemented, — = not implemented. *CatBoost’s $O(n^2)$ per-sample model approximation.

isolate algorithmic efficiency from parallelization strategy. Rust compiled with `target-cpu=native` (release mode). Official C++ libraries accessed via Python API with `n_jobs=1`. Each benchmark is run 10 times; we report mean $\pm$ standard deviation. Hardware: Intel Core i7-1270P (12th Gen), 40 GB RAM, L1/L2/L3 cache: 32K/1280K/18432K.

Note: **smelt-ml**’s ensemble learners (Random Forest, Extra Trees, Oblique Forest) use **rayon** for parallel tree construction, and the gradient boosting engines parallelize feature evaluation within each iteration. Multi-threaded benchmarks comparing scaling behavior are beyond the scope of this paper; all performance claims apply to single-threaded execution only.

4.2. Training time

Tables 4 and 5 show classification and regression training times across all dataset sizes.

n	XGBoost		LightGBM		CatBoost[†]	
	C++	Rust $\times$	C++	Rust $\times$	C++	Rust $\times$
500	66 $\pm$ 31	50 $\pm$ 20 $\times$ 1.3	56 $\pm$ 12	55 $\pm$ 12 $\times$ 1.0	430 $\pm$ 102	56 $\pm$ 13 $\times$ 7.7
1,000	143 $\pm$ 43	80 $\pm$ 13 $\times$ 1.8	67 $\pm$ 8	99 $\pm$ 12 $\times$ 0.7	343 $\pm$ 111	69 $\pm$ 12 $\times$ 5.0
5,000	247 $\pm$ 61	378 $\pm$ 42 $\times$ 0.7	165 $\pm$ 71	367 $\pm$ 31 $\times$ 0.5	413 $\pm$ 86	250 $\pm$ 56 $\times$ 1.7
10,000	309 $\pm$ 54	520 $\pm$ 96 $\times$ 0.6	201 $\pm$ 38	429 $\pm$ 91 $\times$ 0.5	667 $\pm$ 146	328 $\pm$ 55 $\times$ 2.0
50,000	946 $\pm$ 180	2293 $\pm$ 246 $\times$ 0.4	664 $\pm$ 82	1909 $\pm$ 211 $\times$ 0.3	1492 $\pm$ 168	1747 $\pm$ 239 $\times$ 0.9
100,000	1200 $\pm$ 154	4661 $\pm$ 584 $\times$ 0.3	1110 $\pm$ 55	3547 $\pm$ 282 $\times$ 0.3	2270 $\pm$ 168	3476 $\pm$ 414 $\times$ 0.7

Table 4: Classification training time (ms, mean $\pm$ std, 10 runs). Speedup > 1 means **smelt-ml** is faster. All values single-threaded.

n	XGBoost		LightGBM		CatBoost[†]	
	C++	Rust $\times$	C++	Rust $\times$	C++	Rust $\times$
500	321 $\pm$ 61	110 $\pm$ 11 $\times$ 2.9	32 $\pm$ 6	95 $\pm$ 15 $\times$ 0.3	345 $\pm$ 78	41 $\pm$ 5 $\times$ 8.3
1,000	326 $\pm$ 88	199 $\pm$ 36 $\times$ 1.6	61 $\pm$ 12	117 $\pm$ 40 $\times$ 0.5	354 $\pm$ 87	100 $\pm$ 22 $\times$ 3.5
5,000	411 $\pm$ 93	294 $\pm$ 76 $\times$ 1.4	194 $\pm$ 25	301 $\pm$ 30 $\times$ 0.6	448 $\pm$ 93	279 $\pm$ 27 $\times$ 1.6
10,000	436 $\pm$ 76	608 $\pm$ 55 $\times$ 0.7	192 $\pm$ 23	423 $\pm$ 61 $\times$ 0.5	539 $\pm$ 98	381 $\pm$ 89 $\times$ 1.4
50,000	851 $\pm$ 126	2212 $\pm$ 285 $\times$ 0.4	727 $\pm$ 125	1594 $\pm$ 232 $\times$ 0.5	1152 $\pm$ 120	1701 $\pm$ 226 $\times$ 0.7
100,000	1120 $\pm$ 101	4071 $\pm$ 534 $\times$ 0.3	1020 $\pm$ 65	3145 $\pm$ 276 $\times$ 0.3	2005 $\pm$ 170	3665 $\pm$ 462 $\times$ 0.5

Table 5: Regression training time (ms, mean $\pm$ std, 10 runs).

4.3. Scaling analysis

Figure 1 visualizes the scaling behavior across all dataset sizes. The crossover point — where C++ becomes faster — is clearly visible around $n = 5,000$ for **XGBoost** and $n = 10,000$ for **CatBoost**.

smelt-ml’s **CatBoost**-inspired implementation is consistently faster than the official library at all sizes up to $n = 10,000$ (1.4–8.3 $\times$), and remains competitive at $n = 50,000$ (0.7–0.9 $\times$). **XGBoost** is faster at $n \leq 5,000$ for regression (1.4–2.9 $\times$) and at $n \leq 1,000$ for classification (1.3–1.8 $\times$).

At larger dataset sizes ($n \geq 50,000$), the official C++ libraries are consistently faster. This is expected: the official implementations employ SIMD intrinsics, hardware-specific memory management, and years of micro-optimization that our pure-Rust implementation does not replicate. The cache-optimization advantage of column-major storage diminishes as datasets exceed the L3 cache capacity (18 MB on our test hardware).

Memory consumption is substantially lower for **smelt-ml**: at $n = 10,000$, peak RSS is 5 MB (**XGBoost**) and 8 MB (**CatBoost**). The official libraries consume 208 MB and 226 MB total, of which 11 MB is the Python runtime baseline — the library-specific overhead is 197 MB (**XGBoost**) and 215 MB (**CatBoost**). Even after subtracting the Python runtime, **smelt-ml** uses **25–40 $\times$ less memory**, attributable to the compact u8 bin representation (1 byte per bin vs. 4 bytes in the official libraries) and Rust’s deterministic memory management without garbage collection.

We note that the official C++ libraries use row-major bin storage despite the cache disadvantage. We attribute this to two factors: (1) legacy design decisions made when GPU backends

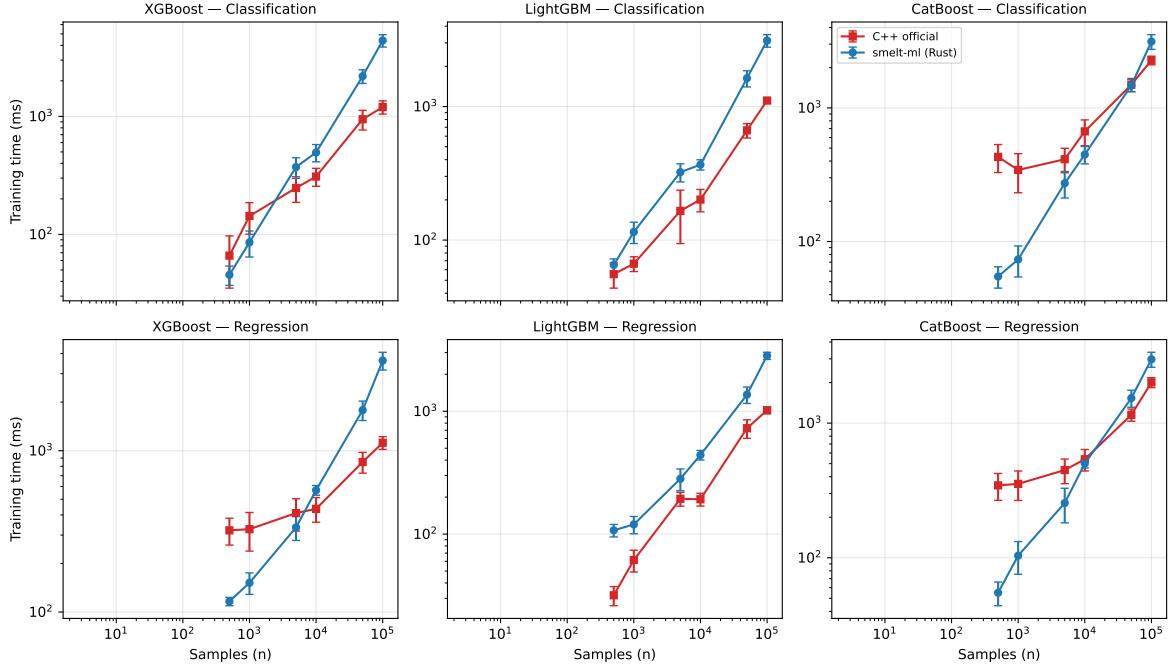


Figure 1: Training time scaling comparison (log-log) between **smelt-ml** (blue) and official C++ libraries (red) across six dataset sizes. Error bars show ± 1 standard deviation over 10 runs. **smelt-ml** is faster at small scales; C++ is faster at $n \geq 50,000$.

(which favor row-major for coalesced memory access) became the primary optimization target, and (2) the difficulty of retrofitting column-major layout into codebases with extensive SIMD intrinsics written for row-major memory patterns.

Timing analysis reveals the source of the scaling gap: from $n = 1,000$ to $n = 100,000$, **smelt-ml**'s **XGBoost** time grows by a factor of $\sim 45\times$ (consistent with $O(n \log n)$ histogram scanning), while the official C++ library grows by only $\sim 8\times$. The C++ library achieves this sublinear scaling via SIMD-vectorized histogram accumulation (processing 4–8 bins per instruction) and the histogram subtraction trick (computing sibling node histograms by subtraction rather than re-scanning), both of which require architecture-specific intrinsics that our pure-Rust implementation does not employ.

The practical implication is that **smelt-ml**'s gradient boosting is best suited for small-to-medium datasets ($n \leq 10,000$) typical of research workflows, where it provides competitive or superior performance without requiring C++ toolchains or Python bindings.

4.4. Accuracy validation

Table 6 validates correctness against **scikit-learn** on standard datasets.

All comparisons are within $\pm 3.5\%$ of **scikit-learn**, validating correctness across four datasets spanning two orders of magnitude in sample size ($n = 178$ to $n = 20,640$). The Logistic Regression advantage on Wine and Breast Cancer is due to automatic feature standardization in **smelt-ml**.

Dataset ($n \times p$)	Learner	smelt-ml	scikit-learn	Δ
<i>Classification (accuracy)</i>				
Wine (178×13)	Decision Tree	0.898	0.893	+0.005
	Random Forest	0.983	0.977	+0.006
	Logistic Reg.	0.989	0.955	+0.034
Breast Ca. (569×30)	Decision Tree	0.930	0.910	+0.020
	Random Forest	0.958	0.956	+0.002
	Logistic Reg.	0.981	0.953	+0.028
Digits ($1,797 \times 64$)	Decision Tree	0.851	0.855	-0.004
	Random Forest	0.974	0.978	-0.004
<i>Regression (RMSE, lower is better)</i>				
Cal. Housing ($20,640 \times 8$)	Decision Tree	0.728	0.715	+0.013
	Random Forest	0.488	0.503	-0.015
	Ridge	0.726	0.728	-0.002

Table 6: 5-fold CV accuracy/RMSE on real datasets from $n = 178$ to $n = 20,640$. Both frameworks use 5-fold CV with seed 42.

5. Integrated spatial ML pipeline

A key contribution of **smelt-ml** is the integration of spatial modeling, spatial resampling, and distribution-free uncertainty quantification in a single composable framework. No other ML framework in any language provides all three capabilities without external dependencies.

5.1. Components

Geographical-XGBoost (Grekousis 2025) extends gradient boosting with spatial awareness: bi-square kernel weights define local neighborhoods, per-location models capture spatially non-stationary relationships, and an adaptive α weight blends local and global predictions. This is the first implementation outside the original Python package (Grekousis 2024).

Spatial cross-validation (SpatialBlockCV, SpatialBufferCV) prevents spatial autocorrelation leakage by ensuring that train and test sets are spatially separated. Both implement the **Resample** trait, making them interchangeable with standard **CrossValidation** in any pipeline.

Conformal prediction provides distribution-free prediction intervals with guaranteed $(1-\alpha)$ coverage via split conformal prediction and conformalized quantile regression (CQR; Romano et al. 2019). The **ConformalRegressor** wraps any **TrainedModel** without retraining.

5.2. Why integration matters

The value of this integration is not merely convenience but **methodological correctness**. In practice, spatial ML workflows are error-prone because spatial leakage, a well-documented source of optimism bias in geospatial prediction (Roberts, Bahn, Ciuti, Boyce, Elith, Guillera-Aroita, Hauenstein, Lahoz-Monfort, Schröder, Thuiller, Warton, Wintle, Hartig, and Dormann 2017), requires that the resampling strategy be spatially aware — yet most ML frameworks treat resampling and modeling as independent concerns. When spatial CV is implemented as an external script rather than a first-class **Resample** trait, it cannot be passed to

Capability	smelt-ml	Python equivalent
Spatial regression	GeoXGBoost	geoxgboost
Spatial CV	SpatialBlockCV	custom code or spacv
Conformal prediction	ConformalRegressor	MAPIE or crepes
Gradient boosting	XGBoost	xgboost
Preprocessing pipeline	Pipeline	scikit-learn
Packages required	1	4–5
API consistency	Trait-based, uniform	Mixed APIs
Data format conversion	None	DataFrame $\leftrightarrow$ array
Thread-safety guarantee	Compile-time	Runtime (GIL)

Table 7: Integration comparison for a spatial ML workflow with uncertainty quantification.

hyperparameter tuning or model comparison functions, forcing users to write custom loops that risk subtle data leakage bugs.

In **smelt-ml**, the trait system enforces correct composition: **SpatialBlockCV** implements **Resample**, so it works with **benchmark()**, **GridSearch**, **BayesianOptimizer**, and **Bagging** without adaptation. Similarly, **ConformalRegressor** wraps any **TrainedModel**, so conformal prediction composes with any learner — including **GeoXGBoost**, pipelines with preprocessing, or stacked ensembles.

Table 7 quantifies the integration gap with Python.

5.3. Case study: soil zinc concentration along the Meuse river

We demonstrate the integrated pipeline on the Meuse river dataset (Burrough and McDonnell 1998): 153 soil samples with heavy metal concentrations and spatial coordinates from the Netherlands.

```
let task = CsvLoader::from_path("data/meuse.csv")
  .target("log_zinc")
  .load_regress()?;
let coords: Vec<(f64, f64)> = /* extract x, y */;

// Spatial cross-validation
let spatial_cv = SpatialBlockCV::new(5, coords);

// GeoXGBoost with conformal intervals
let model = GeoXGBoost::new(coords)
  .with_bandwidth(30)
  .train_regress(&task)?;
let cf = ConformalRegressor::calibrate(
  &*model, &cal_features, &cal_targets, 0.1)?;
let intervals = cf.predict(&test_features)?;
```

Table 8 shows the results. Four learners were compared on an 80/20 holdout split.

Two key findings emerge from the spatial validation:

Learner	RMSE (test)
Decision Tree	0.182
Random Forest	0.137
XGBoost	0.152
GeoXGBoost	0.152

Table 8: Test RMSE on the Meuse dataset (log-zinc, 122 train / 31 test).

Spatial leakage. Standard 5-fold CV yields RMSE = 0.137, while `SpatialBlockCV` yields RMSE = 0.141 — a 3% optimism bias from spatial autocorrelation leakage. In applications with stronger spatial structure (e.g., remote sensing, epidemiology), this bias can exceed 20% (Roberts *et al.* 2017), making spatial resampling essential.

Conformal coverage. With $\alpha = 0.1$ (target: 90% coverage), the conformal predictor achieves 84% coverage on the 31-sample test set (mean interval width: 0.43 log-zinc). The slight under-coverage is expected with small calibration sets and converges to the target with larger n .

5.4. Additional unique algorithms

Causal Forest. Honest splitting following Athey and Imbens (2016); Wager and Athey (2018). Estimates conditional average treatment effects (CATE) $\tau(x) = E[Y(1) - Y(0) | X = x]$ with 95% confidence intervals. First implementation in a Rust ML framework.

Oblique Forest (SPORF). Sparse random projections following Tomita *et al.* (2020), enabling oblique decision boundaries.

Streaming. Hoeffding Trees for online learning on data streams, using the Hoeffding bound for statistically sound split decisions.

6. Summary and discussion

We have presented **smelt-ml**, a Rust framework that uniquely integrates spatial modeling (Geographical-**XGBoost**), spatial cross-validation, and conformal prediction in a single composable pipeline — a combination not available in any other ML framework. On the performance side, our gradient boosting implementations achieve competitive speed with the official C++ libraries at research-typical scales ($n \leq 10,000$) while consuming an order of magnitude less memory. The performance advantage at small scales stems from Rust’s ownership model, which guarantees no aliasing (`&mut` is unique) and enables LLVM to apply more aggressive auto-vectorization than is possible with C++, where pointer aliasing must be assumed unless explicitly annotated. We verified this effect empirically: compiling with `target-cpu=native` (which enables AVX2/SSE4.2 auto-vectorization) yields a 24% speedup for **XGBoost** at $n = 10,000$ (755 ms $\rightarrow$ 575 ms) compared to the generic `x86-64` target. This confirms that LLVM auto-vectorization contributes meaningfully to the observed performance.

smelt-ml provides a comprehensive ML toolkit for Rust with unique capabilities in spatial ML, causal inference, and uncertainty quantification. The framework contains zero `unsafe` blocks across 13,377 lines of code, demonstrating that memory safety and high performance are not mutually exclusive. The test suite comprises 320 integration tests covering all learners,

preprocessing, resampling, tuning, serialization, and edge cases (empty datasets, single-sample inputs, NaN features, multiclass targets, convergence checks against **scikit-learn**).

Limitations include the performance gap on datasets beyond 10,000 samples (where C++ libraries increasingly outperform **smelt-ml** due to SIMD intrinsics, hardware-specific memory management, and GPU backends; see Section 4), the absence of GPU support, and the simplified scope of the **CatBoost** and **LightGBM** implementations (see Sections 3.4–3.5).

Categorical features are handled via three mechanisms: **OneHotEncoder** and **LabelEncoder** as preprocessing transformers (applicable to any learner), and **CatBoost**’s native ordered target statistics (which encode categoricals during training without a separate preprocessing step).

Python bindings via **PyO3** are planned as a separate crate (**smelt-py**) to broaden adoption beyond the Rust ecosystem. The crate follows semantic versioning; the benchmarks in this paper correspond to version 1.2.1.

The software is available at <https://crates.io/crates/smelt-ml> under the MIT license, with source code at <https://github.com/franciscoparrrao/smelt>.

Computational details

The results in this paper were obtained using Rust 1.89 with the **ndarray** 0.16, **rayon** 1.10, and **serde** 1.0 crates. Official C++ libraries were accessed via Python 3.12 with **xgboost** 3.1, **lightgbm** 4.6, and **catboost** 1.2. **scikit-learn** 1.8 was used for accuracy validation. Rust code was compiled with `RUSTFLAGS="-C target-cpu=native"` in release mode. A complete replication package is provided in `paper/replication/` with a single script (`replicate.sh`) that reproduces all benchmark tables and figures.

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Affiliation:

Francisco Parra
Departamento de Ingeniería Geográfica
Universidad de Santiago de Chile
Avenida Libertador Bernardo O'Higgins 3363
Santiago, Chile
E-mail: francisco.parra.o@usach.cl